

Crypt Hw1

Q1:

1.1 One-time pad basics

- A. This type of encryption is mathematically unbreakable because it relies on pure randomness, but it isn't used in practice because of key distribution and authentication issues. Both people must share a massive physical key before they can communicate securely, and if someone is going to have that much security, they may as well transfer the password only. For authentication, there is no verification of who sent the message; as a result, someone may edit the message in transit without the owner's knowledge.
- B. It achieves information-theoretic security since it relies on raw and pure randomness. The key must be generated using physical random processes like thermal noise, instead of a computer algorithm. As a result, no pattern is repeated.

1.2 Security of Cryptographic Function

The birthday attack is a cryptographic method that uses brute force to find two different inputs that yield the same hash output, which would be known as a hash collision. This is done by exploiting the mathematics of the birthday attack. The birthday problem states that in a room of just 23 people, there is a 50% chance that two people share the same birthday, even though there are 365 days in a year. Because the number of pairs grows rapidly as people are added, the chances of finding a collision increase. Therefore, the collision probability increases because finding any collision among random inputs requires fewer attempts than trying to match a specific target, or date.

For an n -bit hash function, the total number of possible output values is $N=2^n$. And the probability $p(k)$ of finding at least one collision after hashing k random inputs is approximately $p(k) \approx 1 - e^{-\frac{k^2}{2 \cdot 2^n}}$. The birthday paradox determines the probability that, out of a group of k randomly chosen elements drawn uniformly from a set of N possible outcomes, at least two elements share the same value. For the approximation, it is easier to calculate the probability that all k hashes are unique, which is denoted as P (no collision). Then, the k -th hash must not match any of the previous $k-1$ hashes. Then, the

Taylor series is used to approximate the exponential function. After that, the final bound formula is calculated as $p(\text{collision}) \approx 1 - e^{(-\frac{k^2}{2^{n-1}})}$. Finally, the 50% threshold is set.

1.3 Kerckhoff's Principle

Kerckhoff's principle in modern cryptography states that a cryptographic system must remain secure even if an attacker knows every detail of the system, except for the secret key. The encryption algorithm, protocol, and hardware should all be public knowledge. And the security depends on hiding the key, not the mechanism, rejecting the "security through obscurity" idea, which states that hiding how a system works makes it safer.

The principle matters in modern cryptography because it allows for peer review, standardization, and practical management. Publishing algorithms allows experts to test for flaws; public algorithms like RSA are public but still secure, and changing a key is simple, whereas changing an algorithm that runs across a whole network is difficult.

1.4 Kerberos and Needham-Schroeder

These protocols are foundational cryptographic network authentication protocols designed to prove identity over insecure networks. Both protocols function as challenge-response mechanisms and ensure that an attacker cannot simply record valid authentication traffic and replay it to gain access.

The protocol requires one party to prove their identity by answering a challenge that is only possible if they possess the secret key. Needham-Schroeder relies on nonces (numbers used once) as its cryptographic challenge. In the Alice-and-Bob example, when Alice wants to talk to Bob, she requests a session key from a trusted key distribution center. Alice can then send the encrypted package to Bob. Bob then sends Alice a challenge to ensure that Alice is who she says that she is. He sends her a nonce. Alice must then decrypt Bob's message, subtract one, for example to make sure the response was not copied, re-encrypt it with the session key, and send it back.

Kerberos modifies the challenge-response dynamic by replacing the nonces with synchronized network timestamps and short-lived authenticators. The challenge is that instead of waiting for a live handshake challenge from the target server, the challenge is driven by time. The response is that when a client requests access to a service, it generates an authenticator containing its identity and the current local timestamp. Encrypted with the session key. The server decrypts this, checks if the timestamp is fresh, and then, for

mutual identification, the server increments the timestamp by one, re-encrypts it, and sends it back.